

# HE2003 Econometrics I

## Tutorial 2

26/08/2024

# Review of Summation Notation

Let  $x_1, \dots, x_n$  denote a sequence of numbers;  $y_1, \dots, y_n$  denote another sequence of numbers; and  $a$  denote a constant. Then, we have:

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n} \Rightarrow \sum_{i=1}^n x_i = n\bar{x}$$

$$\sum_{i=1}^n ax_i = a \sum_{i=1}^n x_i = a \times n\bar{x}$$

$$\sum_{i=1}^n (x_i + y_i) = \sum_{i=1}^n x_i + \sum_{i=1}^n y_i = n\bar{x} + n\bar{y}$$

$$\sum_{i=1}^n a = n \times a$$

# Question 1

Explain the idea of OLS estimation and the related minimization problem.

We consider the following linear regression model:

$$Y_i = \beta_0 + \beta_1 X_i + u_i, i = 1, 2, \dots, n \quad (1)$$

where  $Y_i$  is the dependent variable,  $X_i$  is the explanatory variable, and  $u_i$  is the error term.

$\beta_0 + \beta_1 X$  is the population regression line.  $\beta_0$  is the intercept and  $\beta_1$  is the slope of the population regression line

In practice,  $\beta_0$ ,  $\beta_1$  and the population regression line  $\beta_0 + \beta_1 X$  are unknown. Therefore, we must use data to estimate these unknown values.

# Question 1

The Ordinary Least Squares (OLS) estimator find the estimated regression line as close as possible to the observed data, where closeness is measured by the **sum of squared errors**:

- Here, given some values of  $b_0$  and  $b_1$  (given certain intercept and slope), under the model in (1), the error in predicting  $Y_i$  given  $X_i$  can be written as:  $Y_i - (b_0 + b_1X_i)$ , for each  $i$  (from 1 to  $n$ )
- Therefore, the sum of squared errors can be written as:

$$\sum_{i=1}^n [Y_i - (b_0 + b_1X_i)]^2 = [Y_1 - (b_0 + b_1X_1)]^2 + \dots + [Y_n - (b_0 + b_1X_n)]^2$$

The OLS estimators  $\hat{\beta}_0$  and  $\hat{\beta}_1$ , which are the estimators for the true values  $\beta_0$  and  $\beta_1$ , are obtained by solving the following minimization problem:

$$\min_{b_0, b_1} \sum_{i=1}^n [Y_i - (b_0 + b_1X_i)]^2$$

Intuitively, we search over all possible values of  $b_0$  and  $b_1$ , and find the values that minimize the sum of squared errors. In practice, we solve the first order conditions of the minimization problem.

# Question 2

Derive the formula of the OLS estimators  $\hat{\beta}_0$  and  $\hat{\beta}_1$  following the steps in the lecture note.

The minimization problem:

$$\min_{b_0, b_1} \sum_{i=1}^n [Y_i - (b_0 + b_1 X_i)]^2$$

First, take the partial derivatives with respect to  $b_0$  and  $b_1$ :

$$\frac{\partial}{\partial b_0} \sum_{i=1}^n [Y_i - (b_0 + b_1 X_i)]^2 = -2 \sum_{i=1}^n [Y_i - (b_0 + b_1 X_i)] \quad (2)$$

$$\frac{\partial}{\partial b_1} \sum_{i=1}^n [Y_i - (b_0 + b_1 X_i)]^2 = -2 \sum_{i=1}^n [Y_i - (b_0 + b_1 X_i)] X_i \quad (3)$$

# Question 2

Using First-order condition (FOC) to find the minimum: Find values of  $b_0$  and  $b_1$  for which the first-order derivatives (2) and (3) equal zero. These values are the OLS estimators  $\hat{\beta}_0$  and  $\hat{\beta}_1$ .

$$\begin{aligned}\sum_{i=1}^n [Y_i - (\hat{\beta}_0 + \hat{\beta}_1 X_i)] &= 0 \\ \sum_{i=1}^n Y_i - \sum_{i=1}^n \hat{\beta}_0 - \sum_{i=1}^n \hat{\beta}_1 X_i &= 0 \\ n\bar{Y} - n\hat{\beta}_0 - n\hat{\beta}_1 \bar{X} &= 0 \\ \bar{Y} - \hat{\beta}_0 - \hat{\beta}_1 \bar{X} &= 0 \quad (4)\end{aligned}$$

Where  $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$  and  $\bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i$   
(the sample averages)

$$\begin{aligned}\sum_{i=1}^n [Y_i - (\hat{\beta}_0 + \hat{\beta}_1 X_i)] X_i &= 0 \\ \sum_{i=1}^n X_i Y_i - \sum_{i=1}^n \hat{\beta}_0 X_i - \sum_{i=1}^n \hat{\beta}_1 X_i^2 &= 0 \\ \sum_{i=1}^n X_i Y_i - n\hat{\beta}_0 \bar{X} - \hat{\beta}_1 \sum_{i=1}^n X_i^2 &= 0 \\ \frac{1}{n} \sum_{i=1}^n X_i Y_i - \hat{\beta}_0 \bar{X} - \hat{\beta}_1 \frac{1}{n} \sum_{i=1}^n X_i^2 &= 0 \quad (5)\end{aligned}$$

# Question 2

Solving the two equations (4) and (5), we can obtain the formula of the OLS estimators  $\hat{\beta}_0$  and  $\hat{\beta}_1$ .

Eq. (4) implies  $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$ . Replacing  $\hat{\beta}_0$  in (5) with  $\bar{Y} - \hat{\beta}_1 \bar{X}$ :

$$\frac{1}{n} \sum_{i=1}^n X_i Y_i - (\bar{Y} - \hat{\beta}_1 \bar{X}) \bar{X} - \hat{\beta}_1 \frac{1}{n} \sum_{i=1}^n X_i^2 = 0$$

$$\frac{1}{n} \sum_{i=1}^n X_i Y_i - \bar{X} \bar{Y} + \hat{\beta}_1 \bar{X}^2 - \hat{\beta}_1 \frac{1}{n} \sum_{i=1}^n X_i^2 = 0$$

$$\frac{1}{n} \sum_{i=1}^n X_i Y_i - \bar{X} \bar{Y} = \hat{\beta}_1 \left( \frac{1}{n} \sum_{i=1}^n X_i^2 - \bar{X}^2 \right)$$

$$\Rightarrow \hat{\beta}_1 = \frac{\frac{1}{n} \sum_{i=1}^n X_i Y_i - \bar{X} \bar{Y}}{\frac{1}{n} \sum_{i=1}^n X_i^2 - \bar{X}^2} \quad (6)$$

# Question 2

Thus:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2} = \frac{S_{XY}}{S_X^2}$$

$$\text{and } \hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$$

Where  $S_{XY} = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})$  is the sample covariance between  $X$  and  $Y$  and

$S_X^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$  is the sample variance of  $X$

$\hat{\beta}_0$  is called the estimated intercept, and  $\hat{\beta}_1$  the estimated slope.

They are computed from a sample of  $n$  observations of  $X_i$  and  $Y_i$ ,  $i = 1, \dots, n$



# Question 3

Show the following holds:

$$\frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2} = \frac{\frac{1}{n} \sum_{i=1}^n X_i Y_i - \bar{X} \bar{Y}}{\frac{1}{n} \sum_{i=1}^n X_i^2 - \bar{X}^2}$$

Where  $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$  and  $\bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i$

Hint: show  $\sum_{i=1}^n (X_i - \bar{X})^2 = \sum_{i=1}^n X_i^2 - n\bar{X}^2$  and  $\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y}) = \sum_{i=1}^n X_i Y_i - n\bar{X}\bar{Y}$

# Question 3

$$\begin{aligned}\sum_{i=1}^n (X_i - \bar{X})^2 &= \sum_{i=1}^n (X_i^2 - 2X_i\bar{X} + \bar{X}^2) \\&= \sum_{i=1}^n X_i^2 - \sum_{i=1}^n 2X_i\bar{X} + \sum_{i=1}^n \bar{X}^2 \\&= \sum_{i=1}^n X_i^2 - 2\bar{X} \sum_{i=1}^n X_i + n\bar{X}^2 \\&= \sum_{i=1}^n X_i^2 - 2n\bar{X}^2 + n\bar{X}^2 = \sum_{i=1}^n X_i^2 - n\bar{X}^2 \quad (7)\end{aligned}$$

$$\begin{aligned}\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y}) &= \sum_{i=1}^n (X_i Y_i - X_i \bar{Y} - \bar{X} Y_i + \bar{Y}^2) \\&= \sum_{i=1}^n X_i Y_i - \sum_{i=1}^n X_i \bar{Y} - \sum_{i=1}^n \bar{X} Y_i + \sum_{i=1}^n \bar{Y}^2 \\&= \sum_{i=1}^n X_i Y_i - n\bar{X}\bar{Y} - n\bar{X}\bar{Y} + n\bar{X}\bar{Y} \\&= \sum_{i=1}^n X_i Y_i - n\bar{X}\bar{Y} \quad (8)\end{aligned}$$

# Question 3

Combining (7) and (8), we obtain:

$$\frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2} = \frac{\sum_{i=1}^n X_i Y_i - n\bar{X}\bar{Y}}{\sum_{i=1}^n X_i^2 - n\bar{X}^2}$$

Divide both nominator and denominator by  $n$ :

$$\frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2} = \frac{\frac{1}{n} \sum_{i=1}^n X_i Y_i - \bar{X}\bar{Y}}{\frac{1}{n} \sum_{i=1}^n X_i^2 - \bar{X}^2}$$

# Question 4

Suppose that a researcher, using data on class size ( $CS$ ) and average test scores from 50 third-grade classes, estimate the OLS regression:

$$\widehat{TestScore} = 640.3 - 4.93 \times CS$$

- (a) A classroom has 25 students. What is the OLS regression's prediction for that classroom's average test score?
- (b) Last year a classroom had 21 students, and this year it has 24 students. What is the OLS regression's prediction for the change in the classroom average test score?
- (c) The sample average class size across the 50 classroom is 22.8. What is the sample average of the test scores across the 50 classrooms? (Hint: Review the formulas of the OLS estimators.)

# Question 4

(a) Given that the classroom has 25 students, then the OLS regression's prediction is:

$$\widehat{TestScore} = 640.3 - 4.93 \times 25 = 517.05$$

That is, the OLS regression's prediction for that classroom's average test score is 517.05.

(b) For the case with 21 students (last year):

$$\widehat{TestScore} = 640.3 - 4.93 \times 21 = 536.77$$

For the case with 24 students (this year):

$$\widehat{TestScore} = 640.3 - 4.93 \times 24 = 521.98$$

Therefore, the OLS regression's prediction for the change in the classroom's average test score is  $\Delta \widehat{TestScore} = 521.98 - 536.77 = -14.79$ , where  $\Delta \widehat{TestScore}$  denotes the change in the classroom average test score

# Question 4

(b) Alternatively, according to the interpretation of  $\beta_1 = \frac{\Delta TestScore}{\Delta CS}$ , we have:

$$\widehat{\Delta TestScore} = \hat{\beta}_1 \Delta CS = -4.93 \times (24 - 21) = -14.79$$

Where  $\Delta CS$  denotes the change in the class size.

(c) According to the formulas of the OLS estimators:

$$\begin{aligned}\hat{\beta}_0 &= \bar{Y} - \hat{\beta}_1 \bar{X} \\ \Rightarrow \bar{Y} &= \hat{\beta}_0 + \hat{\beta}_1 \bar{X}\end{aligned}$$

This implies that for the current example, where the sample average of class size is 22.8,  $\hat{\beta}_0 = 640.3$ ,  $\hat{\beta}_1 = -4.93$ , the sample average of the test scores is equal to

$$640.3 - 4.93 \times 22.8 = 527.896$$